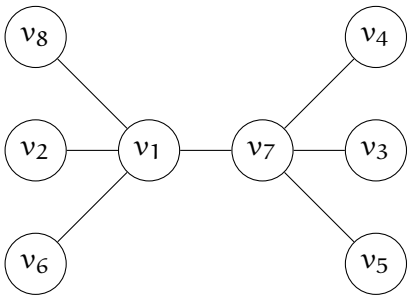
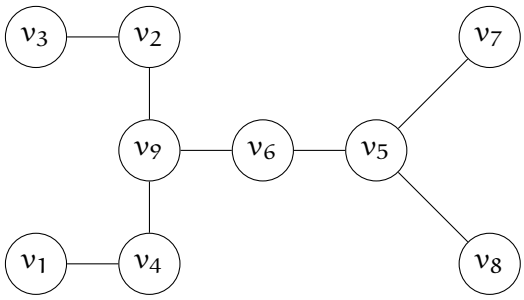
Unless explicitly requested by a problem, do not include sketches as part of your proof. You are free to use the result from any problem on this (or previous) assignment as a part of your solution to a different problem even if you have not solved the former problem.

Problem 1 (0.5 + 0.5 pts)**.** Determine the Prüfer code of the following trees, using the ordering $v_1 \leqslant v_2 \leqslant \dots$ for the vertices.

(a)



(b)



Solution.

(a) $(1, 7, 7, 7, 1, 1)$

(b) $(4, 2, 9, 9, 5, 5, 6)$

□

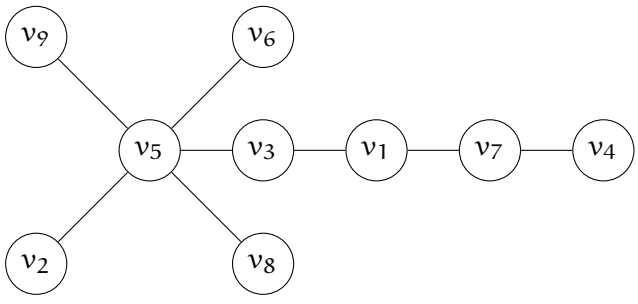
Problem 2 (0.5 + 0.5 pts)**.** For the following sequences, draw a picture of the (labeled) tree which has that sequence as its Prüfer code (using the standard ordering of the integers). Your tree should have vertex-set $\{v_1, \dots, v_n\}$ for some integer n.

(a) $(v_5, v_7, v_5, v_1, v_3, v_5, v_5)$

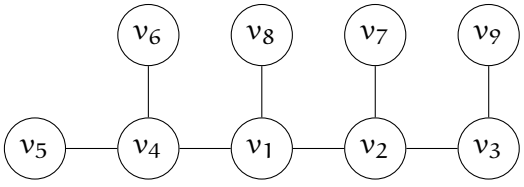
(b) $(v_4, v_4, v_1, v_2, v_1, v_2, v_3)$

Solution.

(a)



(b)



□

Problem 3 (1 pts)**.** Determine (with proof) all trees T (up to isomorphism) on $n \geqslant 2$ vertices whose Prüfer code uses each element of $V(T)$ at most once (under any arbitrary ordering of $V(T)$).

Solution. Under any ordering of $V(T)$, we saw in class that the symbol $x \in V(T)$ appears exactly $\deg_T x - 1$ many times in the Prüfer code of T. Therefore, x appears at most once in the Prüfer code of T if and only if $\deg_T x \leqslant 2$. In other words, the Prüfer code of T sees each element of $V(T)$ at most once if and only if $\Delta(T) \leqslant 2$. This happens if and only if $T \cong P_n$ as per HW4.3. □

Problem 4 (2 pts)**.** Fix an integer $n \geqslant 2$ and let $d_1, \ldots, d_n$ be a sequence of positive integers with $\sum_{i=1}^{n} d_i = 2n - 2$. Use Prüfer codes to show that there is a tree with degree sequence $d_1, \ldots, d_n$.

Solution. Fix a vertex-set $V = \{v_1, \ldots, v_n\}$ with an arbitrary order, say $v_1 < v_2 < \cdots < v_n$. Let $\mathcal{T}_V$ denote the family of all trees on this vertex-set.

Consider the following tuple of elements in V:

$$P = (\underbrace{v_1, \ldots, v_1}_{d_1 - 1 \text{ times}}, \underbrace{v_2, \ldots, v_2}_{d_2 - 1 \text{ times}}, \ldots, \underbrace{v_n, \ldots, v_n}_{d_n - 1 \text{ times}})$$

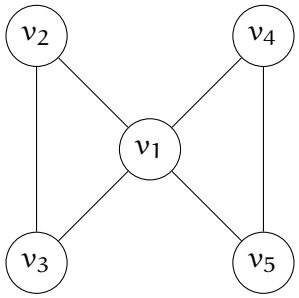
Observe that this makes sense because each $d_i - 1$ is a non-negative integer. Moreover, the number of entries in P is

$$\sum_{i=1}^{n} (d_i - 1) = \sum_{i=1}^{n} d_i - n = 2n - 2 - n = n - 2$$

Thus $P \in V^{n-2}$. Since $\text{Prüfer}_V \colon \mathcal{T}_V \to V^{n-2}$ is a bijection, there is a (unique) tree $T \in \mathcal{T}_V$ such that $\text{Prüfer}_V(T) = P$. We saw in class that $\text{Prüfer}_V(T)$ contains each vertex v_i exactly $\deg_T v_i - 1$ times. Thus $\deg_T v_i - 1 = d_i - 1$, or in other words $\deg_T v_i = d_i$ for all $i \in \{1, \ldots, n\}$. Thus the degree sequence of T is $d_1, \ldots, d_n$. □

Problem 5 (2 pts)**.** For a graph G, define the relation R on $V(G)$ by $u \, R \, v$ if and only if $u = v$ or there is a cycle in G containing both u and v. Find a graph G wherein R is *not* an equivalence relation on $V(G)$. (You are welcome to define G via a picture, though, of course, you must still demonstrate that R is not an equivalence relation on this G)

Solution. Consider the following graph



Observe that $v_2 \, R \, v_1$, as witnessed by the cycle (v_1, v_2, v_3). Similarly, $v_1 \, R \, v_4$, as witnessed by the cycle (v_1, v_4, v_5). However, there is no cycle containing v_2 and v_4, thus v_2 is not related to v_4. Hence R is not transitive, so it is not an equivalence relation. □

Problem 6 (3 pts)**.** For a graph G, define the relation R on $E(G)$ by $e \, R \, s$ if and only if $e = s$ or there is a cycle in G containing both e and s. Prove that R is an equivalence relation on $E(G)$.

Solution. Reflexivity is obvious since $e = e$ for any edge $e \in E(G)$. Symmetry is also obvious since if there is a cycle containing both e and s, then this same cycle contains both s and e.

Transitivity is the only interesting part. Suppose that $e \mathrel{R} s$ and $s \mathrel{R} t$; we must show that $e \mathrel{R} t$. If either $e = s$ or $s = t$, then this is immediate, so we may suppose that e, s, t are three distinct edges.

Thus, let C_{es} be a cycle in G containing both e and s, and let C_{st} be a cycle in G containing both s and t. Label $C_{es} = (v_1, \ldots, v_k)$ where $e = v_1 v_k$ and label $C_{st} = (u_1, \ldots, u_\ell)$ where $t = u_1 u_\ell$; of course, $k, \ell \geqslant 3$. Let i be the smallest index for which $v_i \in \{u_1, \ldots, u_\ell\}$ and let j be the largest index for which $v_j \in \{u_1, \ldots, u_\ell\}$; we begin by showing that both i, j exist and that $i < j$. Indeed, s belongs to both C_{es} and to C_{st} and so $s = v_a v_{a+1}$ and $s = u_b u_{b+1}$ for some $a \in \{1, \ldots, k-1\}$ and some $b \in \{1, \ldots, \ell - 1\}$ (since $e \neq s$ and $t \neq s$). In particular, $v_a, v_{a+1} \in \{u_1, \ldots, u_\ell\}$ and so $1 \leqslant i \leqslant a < a + 1 \leqslant j \leqslant k$. Suppose that $v_i = u_{i'}$ and $v_j = u_{j'}$ for some $i', j' \in \{1, \ldots, \ell\}$; of course $i' \neq j'$. By the definition of i and j, we know that $\{v_1, \ldots, v_{i-1}\} \cap \{u_1, \ldots, u_\ell\} = \{v_k, \ldots, v_{j+1}\} \cap \{u_1, \ldots, u_\ell\} = \varnothing$.

If $i' < j'$, consider the sequence $C = (v_1, \ldots, v_i = u_{i'}, u_{i'-1} \ldots, u_1, u_\ell, u_{\ell-1}, \ldots, u_{j'} = v_j, v_{j+1}, \ldots, v_k)$, and if $i' > j'$, consider the sequence $C = (v_1, \ldots, v_i = u_{i'}, u_{i'+1}, \ldots, u_\ell, u_1, u_2, \ldots, u_{j'} = v_j, v_{j+1}, \ldots, v_k)$. By the earlier remark, in either case, C forms a cycle. Furthermore, since $e = v_1 v_k$ and $t = u_1 u_\ell$, we see that C forms a cycle in G containing both e and t; thus $e \mathrel{R} t$ as needed. □